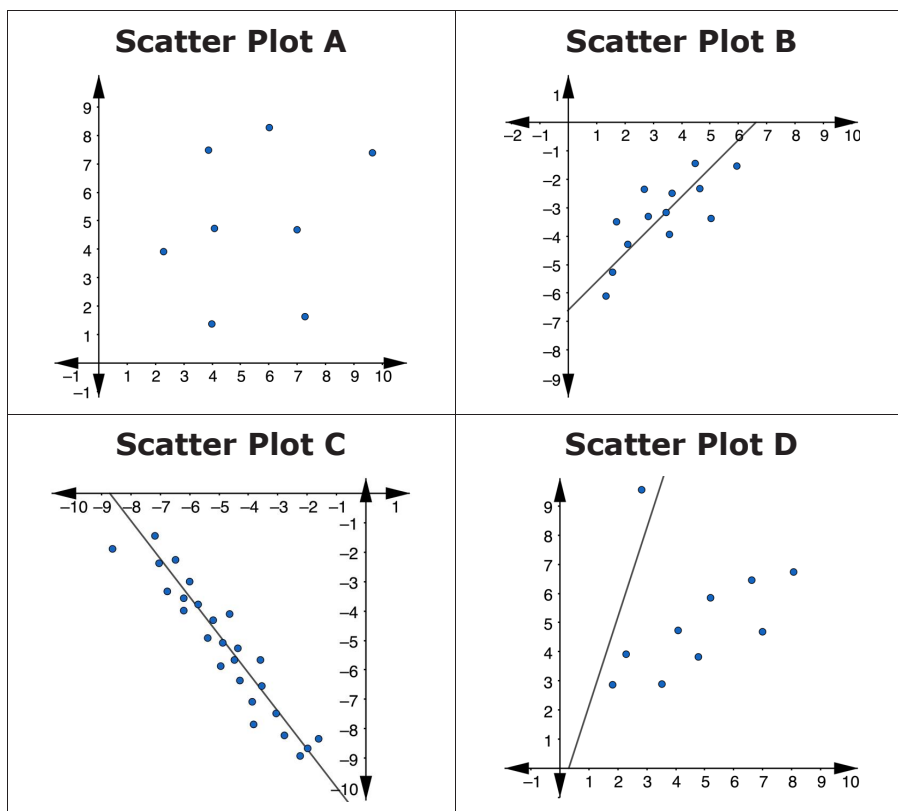




3.5.5 Wrap-Up: Cool Down

Four scatter plots are shown.



1. Complete each statement to explain why one of the scatter plots doesn't belong with the others. Use a different scatter plot to complete each statement.
 - a. Scatter plot ____ doesn't belong because ...
 - b. Scatter plot ____ doesn't belong because ...



Unit 3, Lesson 6: Fitting Linear Functions to Data



Benchmark

MA.912.DP.2.4 Fit a linear function to bivariate numerical data that suggests a linear association and interpret the slope and y -intercept of the model. Use the model to solve real-world problems in terms of the context of the data.



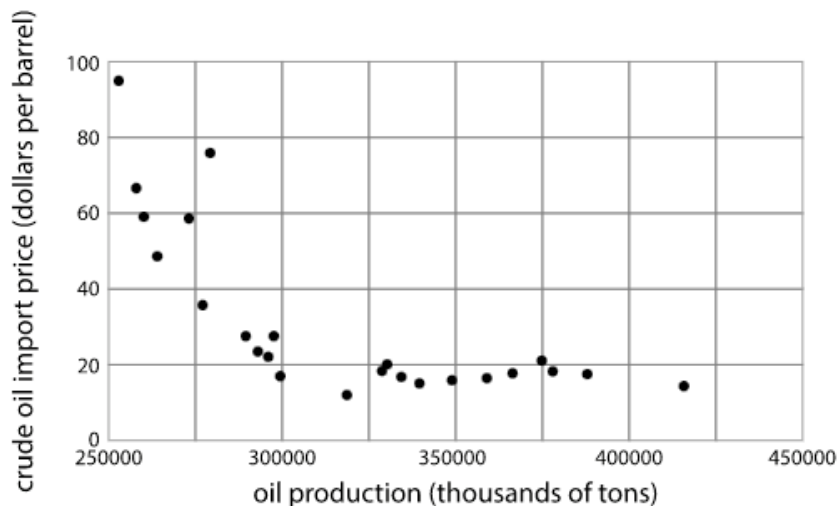
Learning Targets

- ☐ I can formally fit a linear function to bivariate numerical data using technology.
- ☐ I can relate the correlation coefficient to the slope and fit of a linear relationship between two variables.



3.6.1 Warm-Up: Notice and Wonder

The scatter plot shows the relationship between oil production, in thousands of tons, and the crude oil import price, in dollars per barrel.



- Complete the table to indicate one thing you notice and one thing you wonder about when looking at this scatter plot.

One Thing I Notice is ...	One Thing I Wonder is ...



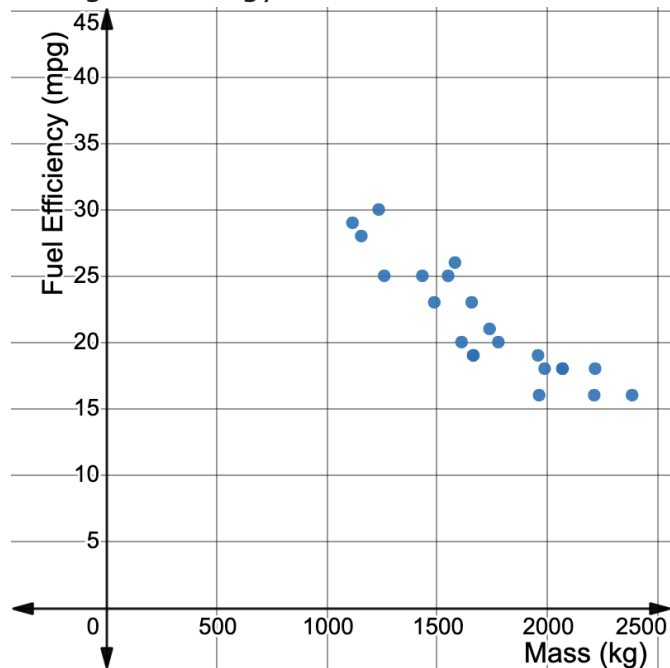
3.6.2 Exploration: Formally Fit a Linear Function to Bivariate Data Using Technology

The mass, in kilograms (kg), and fuel efficiency, in miles per gallon (mpg), of 20 new gas-powered cars is given in the table.

Mass (kg)	Fuel Efficiency (mpg)
1549	25
1610	20
1737	21
1777	20
1155	28
1486	23
1962	16
2384	16
1957	19
2212	16
1115	29
2068	18
1663	19
2068	18
1663	19
2216	18
1432	25
1987	18
1259	25
1580	26
1234	30
1656	23

The raw data can be used to formally fit a line to model the relationship between the data using technology.

1. Tess created the scatter plot shown to represent the data using technology.



Work with your partner to describe the pattern of association between a car's mass and fuel efficiency.

2. Tess fit a line to the data using a graphing calculator. The screenshot shows the results of Tess generating a line, in slope-intercept form, to model the relationship between the mass of a car, x_1 , and the car's fuel efficiency, y_1 .

$$y_1 \sim mx_1 + b$$

STATISTICS

$r^2 = 0.8235$

$r = -0.9075$

PARAMETERS

$m = -0.0109749$ $b = 40.2965$

RESIDUALS

e_1

- a. The "~" symbol is used to mean one quantity is "approximately equivalent to" another quantity. Discuss with your partner why you think this symbol would be used rather than an "=" when determining a line of best fit. Summarize your discussion.
 - b. According to the screenshot, what are the slope and y-intercept of the line of best fit generated using technology, rounded to the nearest thousandth?
- | | |
|--------------------|--|
| slope | |
| y-intercept | |
- c. Write the slope-intercept form of the equation of the line of best fit Tess generated with technology.
 - d. Interpret the slope in the given context.
 - e. Interpret the y-intercept in the given context.

- f. Explain whether the y -intercept makes sense in this context.

Extrapolation occurs when a model is used to make predictions for x -values that fall outside of the data set.

Problems may occur when extrapolating, since extending the trend line far past the x -values of the data set may make the predictions less reliable.

3. Explain why it would not be reliable to use Tess' model for this data set to make predictions about the fuel efficiency of hybrid or electric vehicles.

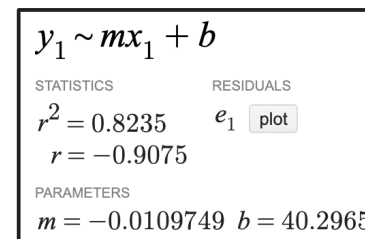
4. Use inequalities to express an appropriate domain and range for making predictions using the linear equation modeling the data.

domain	
range	



3.6.3 Guided Instruction: Relating the Correlation Coefficient, r , to the Slope of the Fit of the Regression Line

The screenshot of the line Tess formally fit to the data using technology included a statistic, r .



The statistic, r , is the **correlation coefficient**. It measures the strength and direction of the linear association between the variables in a set of bivariate numerical data.



The **correlation coefficient**, r , is a unitless value between -1 and 1 .

- A correlation coefficient of 1 represents a perfect linear relationship with a positive direction.
- A correlation coefficient of -1 represents a perfect linear relationship with a negative direction.
- The closer the correlation coefficient is to 0 , the weaker the linear association between the variables.

1. Complete the statement to interpret the correlation coefficient for Tess' linear model.

The correlation coefficient -0.9075 means there is a

☐ weak
☐ strong

☐ linear
☐ nonlinear

association between a car's

mass and its fuel efficiency because the value of r is

close to

☐ 0.
☐ -1 or 1 .

The sign of r is

☐ positive,
☐ negative,

which means the direction of the association between

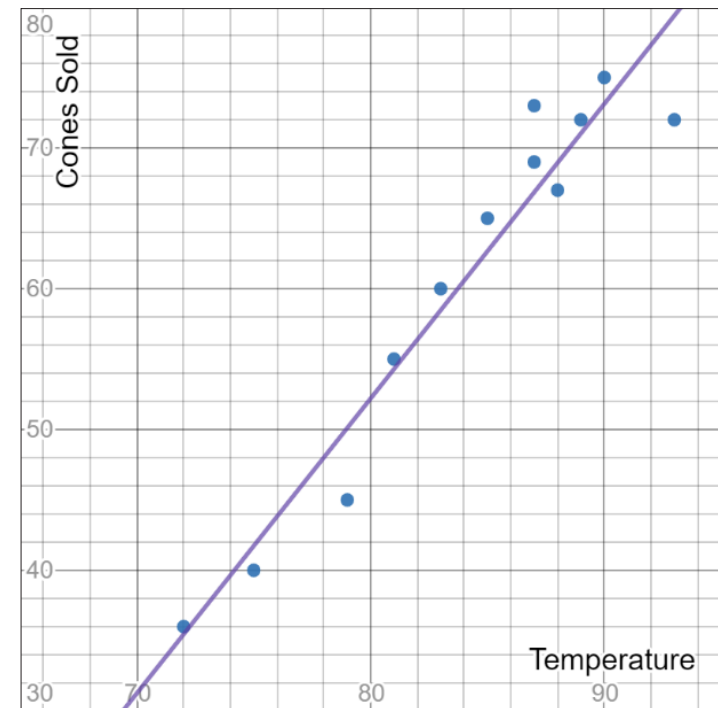
the variables is

☐ positive.
☐ negative.

2. Albert, an ice cream vendor at Jones Beach, records the number of cones he sells each day, as well as the daily high temperature. The table and the scatter plot below show his data for 12 days in June.

Relationship Between Temperature and Cones Sold

Temperature ($^{\circ}F$)	81	72	88	85	89	90
Number of Cones Sold	55	36	67	65	72	75
Temperature ($^{\circ}F$)	87	75	83	79	87	93
Number of Cones Sold	73	40	60	45	69	72



The equation of the line of best fit for this data is $y = 2.0925x - 115.195$ with a correlation coefficient $r = 0.9646$.

Complete the statements.

There is a

☐ weak,
☐ strong,

☐ positive,
☐ negative,

☐ linear
☐ nonlinear

association between the temperature outside and the

number of ice cream cones Albert sells in June. The

number of cones he sells

☐ increases
☐ decreases

by

approximately

☐ 2
☐ 115

for every 1°F increase in the

temperature outside.



3.6.4 Guided Practice: Formally Fitting a Line of Best Fit to Data and Using It to Solve Problems

- Kara wants to sell her Subaru Forester, but does not know what the listing price should be. She checks on a popular used-car site and finds 16 Subaru Foresters listed. Kara uses the data to construct two different scatter plots – one that relates the vehicle's age, in years, to the sale price, and the other that relates the number of miles on each vehicle to its sale price.

Two different scatter plots with lines of best fit that were generated using technology are shown. Scatter Plot A relates the mileage of each vehicle to the sale price. Scatter Plot B relates the age of each vehicle to the sale price.

Scatter Plot A	Scatter Plot B
$y_1 \sim mx_1 + b$ STATISTICS $r^2 = 0.7955$ $r = -0.8919$ PARAMETERS $m = -0.12654$	$y_1 \sim mx_1 + b$ STATISTICS $r^2 = 0.8209$ $r = -0.906$ PARAMETERS $m = -1451.95$
RESIDUALS e_1 plot $b = 23837.8$	RESIDUALS e_1 plot $b = 23746.4$

- a. Write the equation for each line of best fit using the values generated with technology.

Line of Best Fit for Scatter Plot A	Line of Best Fit for Scatter Plot B

- b. Interpret the slope of the line of best fit for Scatter Plot B.
- c. Describe the strength and direction of the linear association between mileage and sale price of a used Subaru Forester.

2. Suppose Kara's Subaru Forester is 9 years old.
 - a. Which equation could she use to predict a good sale price for her car?
 - b. What sale price should Kara list her car for based on its age?



3.6.5 Wrap-Up: Reflection

1. Complete each statement.
 - a. After today's lesson, I am still unsure about ...
 - b. To fill in this gap, I intend to ...